# TOYJOY

## Toy Retailer Company

## Inventory Optimization & Demand Forecasting Case Study

A Comprehensive Data Analysis Project
For Junior Data Analysts

**Prepared by:** Junior Data Analyst Team
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**Period Covered:** April 2023 – March 2026 (3 Years)
**Report Date:** July 2026

# 1. Executive Summary

ToyJoy is a leading toy retailer operating through both physical retail stores and online sales channels. Over the past three years, the company has experienced significant fluctuations in inventory levels due to changing customer demand, seasonal purchasing behavior, and inconsistent replenishment planning. While several products frequently sold out during peak seasons, others accumulated excessive inventory, resulting in increased warehouse costs and reduced working capital efficiency.

This case study documents the complete analytical journey undertaken to evaluate ToyJoy's inventory performance using Power BI. The study addresses five key business questions prioritized by management:

- Identify products and categories carrying excess inventory.
- Determine products experiencing frequent stockouts.
- Analyze seasonal demand trends across the three-year period.
- Recommend inventory adjustments for slow-moving and high-demand products.
- Optimize reorder points and replenishment quantities to improve inventory turnover and reduce carrying costs.

The analysis was conducted using Power BI and included campaign performance, engagement metrics, conversion efficiency, campaign duration analysis, and channel-level profitability.

### Key Findings at a Glance

- Total Revenue generated during the analysis period was approximately ₹32.13 Million.
- A total of 53,601 units were sold across all products over the three-year analysis period.
- The average inventory level was approximately 111 units per product-store combination.
- Excess Inventory Value was approximately ₹207.70 Million, indicating substantial capital tied up in slow-moving inventory.
- Average Inventory Turnover was 48.50, with Princess Doll recording the highest turnover (63.32) and Chess Master the lowest (36.91), indicating opportunities for product-specific inventory optimization.
- Stockout Rate remained relatively low at 0.17%, though certain products experienced repeated stockout events.
- Monopoly Classic holds the highest inventory investment (₹37.43 Million), followed by Robo Defender (₹29.91 Million), indicating that a significant portion of working capital is concentrated in a few products and should be regularly reviewed to minimize carrying costs.
- Seasonal demand spikes were observed during holiday and festival periods, requiring proactive inventory planning.

The report concludes with practical recommendations covering inventory optimization, seasonal forecasting, replenishment planning, and long-term inventory management strategies designed to improve operational efficiency and cash flow.

# 2. Business Scenario & Background

## 2.1 Company Profile

| Parameter | Details |
|---|---|
| **Company Name** | ToyJoy |
| **Industry** | Toys & Games Retail |
| **Business Model** | Retail Stores & Online Sales |
| **Product Categories** | Action Figures, Board Games, Dolls, Puzzles |
| **Analysis Period** | April 2023 – March 2026 |
| **Reporting Currency** | Indian Rupee (INR) |
| **Sales Channels** | Retail Stores and Online |

## 2.2 Problem Statement

After the holiday season, ToyJoy management identified several inventory-related challenges:

- Excess inventory occupying warehouse space and increasing holding costs.
- Frequent stockouts of popular products leading to lost sales.
- Difficulty predicting seasonal demand fluctuations.
- Limited warehouse capacity requiring more efficient inventory allocation.
- Need for data-driven reorder planning and inventory optimization.

The primary objective of this analysis is to optimize inventory planning in order to reduce overstock, minimize stockouts, and improve overall cash flow efficiency.

# 3. Key Performance Indicators (KPIs)

The following KPIs were selected to directly map to the business objectives. Each KPI has a defined formula and a target benchmark established by management.

| KPI | Formula | Business Purpose |
|---|---|---|
| **Units Sold per Product** | SUM(Units Sold) | Measures the total quantity of products sold over the analysis period to identify product demand. |
| **Inventory Turnover Rate** | Units Sold ÷ Average Inventory | Evaluates how efficiently inventory is sold and replenished; higher turnover indicates better inventory management. |
| **Stockout Rate** | (Number of Stockout Records ÷ Total Inventory Records) × 100 | Measures the frequency of stockout occurrences to identify products causing lost sales. |
| **Excess Inventory Value** | SUM(Excess Inventory × Unit Cost) | Quantifies the monetary value of overstock inventory tying up working capital. |
| **Seasonal Demand Pattern** | Monthly & Seasonal Trend Analysis of Units Sold / Revenue | Identifies demand fluctuations across seasons to support inventory planning and forecasting. |
| **Revenue Contribution** | SUM(Revenue) | Measures the revenue generated by each product or category to identify key revenue drivers. |
| **Average Inventory Value** | SUM(Inventory Value) ÷ Number of Inventory Records | Indicates the average monetary value of inventory held during the analysis period for inventory investment assessment. |

# 4. Dataset Overview

## 4.1 Dataset Specification

The dataset `toyjoy_inventory.csv` was created to simulate inventory and sales transactions for ToyJoy over a three-year period (April 2023–March 2026) across different stores, product category, and products. It contains approximately 5,500 rows of records covering multiple products, categories, stores, and seasonal periods.

| Column Name | Descriptions & Notes |
|---|---|
| **ProductID** | Unique identifier assigned to each toy product. |
| **ProductName** | Name of the toy product. |
| **Category** | Product category (Action Figures, Board Games, Dolls, Puzzles); minor naming inconsistencies corrected during cleaning |
| **StoreID** | Unique identifier for each retail store. |
| **Date** | Transaction date covering April 2023 to March 2026. |
| **UnitsSold** | Number of units sold during the reporting period. |
| **UnitsInStock** | Remaining inventory available after sales; ~1.9% missing values in the raw dataset |
| **UnitCost** | Cost price per unit used for inventory valuation. |
| **UnitPrice** | Selling price per unit used to calculate revenue. |
| **Season** | Business season used for seasonal demand analysis. |

## 4.2 Data Quality Assessment

The dataset was reviewed thoroughly before analysis to ensure consistency and reliability.

### Data Cleaning & Validation Performed:

- **Duplicate Records:** Removed 170 duplicate rows based on ProductID + StoreID + Date, retaining the latest valid record.

- **Missing Values:** UnitsInStock (103 nulls) imputed using the product-wise median inventory.
- **Zero Inventory:** Retained 9 zero inventory records as genuine stockout scenarios.
- **Category Names:** Standardized inconsistent category names for uniform reporting.
- **Date Validation:** Verified all records fall within Apr 2023 – Mar 2026.
- **Numeric Fields:** Validated UnitsSold, UnitsInStock, UnitCost, and UnitPrice for consistency.

# 5. Business Questions & Analytical Approach

## Q1. Which products or categories have excess inventory that ties up capital?

**Approach:**

- Created an Overstock Flag to identify products where Units in Stock exceeded twice the Units Sold.
- Calculated Excess Inventory Value only for products flagged as overstocked using their inventory value.
- Aggregated Excess Inventory Value by product and category to identify inventory tying up the highest working capital.

**Key Finding:**
Monopoly Classic carries the highest excess inventory value (₹37.43 Million), followed by Robo Defender (₹29.91 Million) and Baby Doll Deluxe (₹24.33 Million). These products represent the largest share of working capital tied up in inventory and should be prioritized for inventory optimization.

## Q2. Which products are frequently out of stock, causing lost sales?

**Approach:**

- Calculated Stockout Count for each product.
- Compared stockout rate with Inventory Turnover (i.e. ratio of total units sold to average inventory).

**Key Finding:**

Princess Doll experienced the highest stockout frequency (5 events), followed by Baby Doll Deluxe and SuperHero Warrior (2 events each). These recurring stockouts indicate the need for improved replenishment planning and higher safety stock levels.

## Q3. Can you identify seasonal trends in sales to guide stocking for the next period?

**Approach:**

- Aggregated Units Sold by business season.
- Compared seasonal demand across all six business seasons.

**Key Finding:**

The Festive Season recorded the highest demand (19,537 units sold), followed by Summer Vacation (11,562 units) and Regular Season (11,150 units). These recurring seasonal trends support proactive inventory planning before high-demand periods.

## Q4. Recommend inventory adjustments for underperforming and high-demand products.

**Approach:**

- Compared Units Sold, Total Revenue, and Total Units in Stock for each product.
- Used the scatter plot to identify products with high demand, low inventory availability, and relatively lower-performing products.
- Recommended inventory adjustments based on product demand and inventory position.

**Key Finding:**

Robo Defender (6,273 units; ₹5.01M), Baby Doll Deluxe (6,845 units; ₹4.44M), SuperHero Warrior (5,832 units; ₹4.08M), and Princess Doll (6,731 units; ₹3.70M) demonstrate consistently high demand and should receive replenishment priority. Conversely, Family Ludo (4,620 units; ₹1.38M) and Animal Puzzle (4,688 units; ₹1.87M) show comparatively lower demand and should be considered for reduced reorder quantities. Monopoly Classic requires close inventory monitoring due to its high inventory investment despite moderate sales volume (4,800 units).

## Q5. Suggest strategies to optimize reorder points and quantities to reduce overstock and stockouts.

**Approach:**

- Compared Inventory Turnover, Stockout Count, Units Sold, and Excess Inventory Value for each product.
- Evaluated product demand using the Demand vs Inventory Position analysis.

- Identified products requiring more frequent replenishment and products requiring reduced reorder quantities.

**Key Finding:**

Products with high inventory turnover and recurring stockouts, such as Princess Doll (63.32 turnover, 5 stockouts), Baby Doll Deluxe (62.71, 2 stockouts), and SuperHero Warrior (52.49, 2 stockouts), require higher reorder points and more frequent replenishment. Conversely, products with lower inventory turnover, including Chess Master (36.91) and Family Ludo (40.48), should follow lower reorder quantities to minimize excess inventory. Additionally, Monopoly Classic should maintain controlled replenishment because it holds the highest Excess Inventory Value (₹37.43M) despite a comparatively lower inventory turnover (42.56).

# 6. Exploratory Data Analysis (EDA) Guidance

## 6.1 Summary Statistics to Report

| Metric | Value |
|---|---|
| **Total Products** | 10 |
| **Analysis Period** | April 2023 – March 2026 (3 Years) |
| **Product Categories** | 4 (Action Figures, Board Games, Dolls, Puzzles) |
| **Total Revenue** | ₹32,132,799 (₹32.13 Million) |
| **Total Units Sold** | 53,601 Units |
| **Average Inventory** | 110.83 Units |
| **Average Inventory Value** | ₹39,415.94 |
| **Average Inventory Turnover** | 48.50 |
| **Excess Inventory Value** | ₹207,695,840 (₹207.70 Million) |
| **Stockout Rate** | 0.17% |

## 6.2 Recommended Visualizations

| Visualization (Chart Title in Dashboard) | What it Answers |
|---|---|

| | |
|---|---|
| **KPI Cards: Executive KPIs** | What is the current overall inventory performance of ToyJoy? |
| **Bar Chart: Inventory Value by Category** | Which product categories account for the highest inventory investment? |
| **Bar Chart: Top 10 Overstock Products** | Which products have excess inventory tying up working capital? |
| **Matrix: Product Performance Summary** | Which products have high inventory turnover, stockout rates, and inventory value? |
| **Bar Chart: Top Stockout Products** | Which products experience recurring stockouts and potentially lost sales? |
| **Line Chart: Monthly Product Sales Trend** | How do product sales vary across different months? |
| **Column Chart: Sales by Season** | Which seasons generate the highest product demand? |
| **Scatter Plot: Demand vs Inventory Position** | Is inventory investment aligned with product demand and revenue? |
| **Dual Line Chart: Inventory Investment vs Revenue Trend** | How do inventory investment and revenue change over time? |

## 6.3 Data Cleaning Steps

1. Imported the **toyjoy_inventory.csv** dataset into Power BI and performed an initial data inspection.
2. Cleaned and trimmed text fields to remove leading and trailing spaces, improving data consistency.
3. Standardized product category names to eliminate naming inconsistencies across the dataset.
4. Removed duplicate records by treating each **Product ID – Store ID – Date** combination as a unique inventory record. Where duplicates existed, the latest record was retained.
5. Replaced missing values in the **Inventory** column with the median inventory level while retaining zero inventory values as genuine stockout scenarios.
6. Created relationships between the **Inventory**, **Product Category**, and **Date–Season** tables to support category-wise and seasonal analysis.
7. Created calculated columns, DAX measures, and KPIs including **Units Sold, Revenue, Inventory Turnover Rate, Stockout Rate, Average Inventory Value, Excess Inventory Value,** and **Revenue Contribution**.
8. Validated all calculated measures and relationships before developing the final Power BI dashboard.

### Observations:

- Product category names were standardized, resulting in consistent category-level reporting.
- Duplicate inventory records were successfully removed without affecting historical inventory trends.
- Median imputation minimized the impact of missing inventory values while avoiding bias caused by extreme observations.
- Zero inventory values were preserved as valid stockout events for accurate stockout rate calculations.
- The processed dataset was suitable for inventory turnover analysis, seasonal demand analysis, stockout identification, and revenue contribution analysis

# 7. Expected Insights & Hidden Patterns

## 7.1 Inventory Performance Insights

### Product Performance

- Robo Defender (₹5.01M), Monopoly Classic (₹4.80M), Baby Doll Deluxe (₹4.44M), and SuperHero Warrior (₹4.08M) are the highest revenue-generating products, making them the primary contributors to ToyJoy's sales.
- Monopoly Classic (₹37.43M) holds the largest inventory investment, followed by Robo Defender (₹29.91M) and Baby Doll Deluxe (₹24.33M), indicating significant working capital tied up in these products.
- Princess Doll (63.32) and Baby Doll Deluxe (62.71) achieved the highest Inventory Turnover Rates, while Chess Master (36.91) recorded the lowest turnover, suggesting comparatively slower inventory movement.
- Revenue generation is concentrated among a limited number of products, highlighting the need to prioritize inventory planning for high-performing SKUs.

## 7.2 Inventory Availability & Replenishment Insights

### Stock Optimization

- Princess Doll experienced the highest stockout frequency (5 events), followed by Baby Doll Deluxe and SuperHero Warrior (2 events each), indicating recurring supply shortages.
- Stockouts occurred primarily among high-demand products, suggesting current reorder levels are insufficient during peak demand.
- Inventory Turnover varies from 36.91 to 63.32, demonstrating that replenishment strategies should be product-specific rather than uniform across all SKUs.
- Reducing inventory for slow-moving products while increasing safety stock for fast-moving items can improve inventory utilization and customer service levels.

## 7.3 Seasonality & Demand Insights

### Seasonal Sales Pattern

- Festive Season generated the highest demand with 19,537 units sold, followed by Summer Vacation (11,562) and Regular Season (11,150).
- Back to School (3,721), Financial Year End (3,810), and Christmas & New Year (3,821) contributed comparatively lower sales volumes.
- Demand during the Festive Season was approximately 69% higher than Summer Vacation, highlighting the need for proactive inventory build-up before major festivals.
- Historical seasonal trends provide a reliable basis for improving demand forecasting and replenishment planning.

# 8. Recommendations

## 8.1 Quick Wins (This Quarter)

- **Reduce inventory for slow-moving products:** Review products such as Chess Master and Family Ludo, which recorded the lowest inventory turnover, and reduce future purchase quantities to minimize excess inventory and carrying costs.
- **Increase safety stock for high-demand products:** Products including Princess Doll, Baby Doll Deluxe, and SuperHero Warrior experienced recurring stockouts. Increasing safety stock before peak demand periods can improve product availability and reduce lost sales.
- **Prioritize inventory investment in top-performing products:** Maintain adequate inventory for Robo Defender, Monopoly Classic, Baby Doll Deluxe, and SuperHero Warrior, as they generate the highest revenue contribution.
- **Prepare inventory for peak seasons:** Increase procurement before the Festive Season and Summer Vacation, which recorded the highest sales volumes, while maintaining leaner inventory during lower-demand periods.
- **Monitor excess inventory monthly:** Regularly review products with high inventory value but moderate sales to identify opportunities for stock redistribution or promotional clearance.

## 8.2 Long-Term Strategies (Next 2-4 Quarters)

- **Implement seasonal demand forecasting:** Use the three years of historical sales and seasonal patterns to forecast inventory requirements and improve procurement planning.
- **Adopt dynamic reorder policies:** Base reorder quantities on historical sales velocity, inventory turnover, and seasonal demand rather than fixed replenishment levels.
- **Introduce ABC inventory classification:** Classify products according to revenue contribution and inventory value to prioritize inventory investment in high-impact products.
- **Review product portfolio performance periodically:** Evaluate slow-moving products regularly to optimize assortment, reduce warehouse utilization, and improve overall inventory efficiency.

## 8.3 Inventory Optimization Framework

| Focus Area | Recommended Strategy |
|---|---|
| **High-Revenue Products** | Maintain higher inventory availability and prioritize replenishment to prevent lost sales. |
| **High Stockout Products** | Increase safety stock and review reorder thresholds during peak demand periods. |
| **Slow-Moving Products** | Reduce replenishment quantities and use targeted promotions to improve inventory turnover. |
| **Procurement Planning** | Use demand forecasts to align purchasing decisions with expected sales patterns. |

# 9. Constraints & Assumptions

| Constraint / Assumption | Implication for Analysis |
|---|---|
| **Data covers the last three years** | Suitable for identifying historical demand patterns and seasonality but may not capture long-term market changes beyond the available period. |
| **Product ID + Store ID + Date represents a unique inventory record** | Duplicate records were treated as data inconsistencies, and the latest available record was retained. |
| **Missing inventory values were replaced with the median inventory level** | Median imputation reduces the influence of extreme values while preserving the overall inventory distribution. |
| **Zero inventory values were retained** | Zero inventory was assumed to represent genuine stockout scenarios rather than missing data. |
| **Products were classified as overstocked when Units in Stock exceeded twice the Units Sold** | Since the dataset did not define an overstock threshold, this business rule was assumed to identify products with inventory levels significantly exceeding historical demand. This assumption was used to calculate the **Overstock Flag** and **Excess Inventory Value**. |

| | |
|---|---|
| **Category names were standardized during preprocessing** | Ensures consistent category-level aggregation and reporting. |
| **Warehouse capacity and procurement lead times were not explicitly modeled** | Recommendations focus on inventory optimization based on historical sales and inventory data rather than operational constraints. |